

PART-A DICTIONARIES

Q1. Create an empty Dictionary

```
inventory = {}  
  
print(inventory)  
  
{}
```

Q2. Store the first product details in variables

```
ProductNo1_name = "Mobile Phone"  
ProductNo1_quantity = 5  
ProductNo1_price = 20000  
ProductNo1_releaseYear = 2020  
  
print(ProductNo1_name)  
print(ProductNo1_quantity)  
print(ProductNo1_price)  
print(ProductNo1_releaseYear)  
  
Mobile Phone  
5  
20000  
2020
```

Q3. Add details in inventory

```
inventory["ProductNo1"] = {  
    "Name": ProductNo1_name,  
    "Quantity": ProductNo1_quantity,  
    "Price": ProductNo1_price,  
    "ReleaseYear": ProductNo1_releaseYear  
}  
  
print(inventory)  
  
{'ProductNo1': {'Name': 'Mobile Phone', 'Quantity': 5, 'Price': 20000,  
'ReleaseYear': 2020}}
```

Q4. Store the second product details in variables

```
ProductNo2_name = "Laptop"  
ProductNo2_quantity = 3  
ProductNo2_price = 50000  
ProductNo2_releaseYear = 2022
```

```
print(ProductNo2_name)
print(ProductNo2_quantity)
print(ProductNo2_price)
print(ProductNo2_releaseYear)
```

```
Laptop
3
50000
2022
```

Q5. Add the second product details in inventory

```
inventory["ProductNo2"] = {
    "Name": ProductNo2_name,
    "Quantity": ProductNo2_quantity,
    "Price": ProductNo2_price,
    "ReleaseYear": ProductNo2_releaseYear
}

print(inventory)

{'ProductNo1': {'Name': 'Mobile Phone', 'Quantity': 5, 'Price': 20000, 'ReleaseYear': 2020}, 'ProductNo2': {'Name': 'Laptop', 'Quantity': 3, 'Price': 50000, 'ReleaseYear': 2022}}
```

Q6. Display the products present in inventory

```
print("Products present in inventory:")
print(inventory)

Products present in inventory:
{'ProductNo1': {'Name': 'Mobile Phone', 'Quantity': 5, 'Price': 20000, 'ReleaseYear': 2020}, 'ProductNo2': {'Name': 'Laptop', 'Quantity': 3, 'Price': 50000, 'ReleaseYear': 2022}}
```

Q7. Check if ProductNo1_releaseYear and ProductNo2_releaseYear are in inventory

```
print(ProductNo1_releaseYear in inventory["ProductNo1"].values())
print(ProductNo2_releaseYear in inventory["ProductNo2"].values())

True
True
```

Q8. Delete release year of both products

```
del inventory["ProductNo1"]["ReleaseYear"]
del inventory["ProductNo2"]["ReleaseYear"]
```

```
print(inventory)

{'ProductNo1': {'Name': 'Mobile Phone', 'Quantity': 5, 'Price': 20000}, 'ProductNo2': {'Name': 'Laptop', 'Quantity': 3, 'Price': 50000}}
```

PART B — TUPLES

Q1. Create a tuple called prices

```
prices = (250, 300, 150, 400, 100, 350, 200)

print(prices)

(250, 300, 150, 400, 100, 350, 200)
```

Q2. Find and print the highest price

```
print("Highest price:", max(prices))

Highest price: 400
```

Q3. Find and print the lowest price

```
print("Lowest price:", min(prices))

Lowest price: 100
```

Q4. Calculate and print the total sum

```
print("Total sum:", sum(prices))

Total sum: 1750
```

Q5. Convert the tuple into a sorted list

```
sorted_prices = sorted(prices)

print("Sorted list:", sorted_prices)

Sorted list: [100, 150, 200, 250, 300, 350, 400]
```

Q6. Try to modify an element in the tuple

```
prices[0] = 500
```

```
-----  
-----  
TypeError                                Traceback (most recent call  
last)  
/tmp/ipykernel_860/671771468.py in <cell line: 0>()  
----> 1 prices[0] = 500  
  
TypeError: 'tuple' object does not support item assignment
```

EXPLANATION - A tuple is immutable, so its elements cannot be modified after creation.